EC601 — Phase 1 Assignment: Build a Tutorial with an LLM

Duration: 1 week **Submit:** Tutorial + integrated quizzes + your LLM conversation log + short reflection

The Assignment

Pick one research area from the Areas & Faculty list. Your job this week:

1. **Read the foundations.** Find and read 3–5 key papers or authoritative resources in the area (recent survey papers, seminal papers, or the listed BU faculty's recent work).
2. **Work with an LLM (Claude or ChatGPT) as your co-author** to create a strong tutorial that teaches the area to a fellow masters student who has never seen it.
3. **Integrate quizzes.** Use the LLM to generate quiz questions embedded throughout the tutorial — not just at the end. Each major section should have 2–4 questions (multiple choice or short answer) with answers and explanations.
4. **Verify everything.** You are responsible for the accuracy of the tutorial. LLMs make mistakes and invent citations. Every factual claim and every reference must be checked against the actual papers. An unverified hallucinated citation is an automatic major deduction.

Deliverables

1. **The tutorial** (PDF, Markdown, or web page) — roughly 8–15 pages equivalent:
    - Motivation: why this area matters, real applications
    - Core concepts explained from first principles
    - Current state of the art: what the key papers do, what's still open
    - While working with the LLM,request to verify your understanding via questions. Submit a link to the chat.
    - A properly formatted reference list (verified — every reference must be real and must say what you claim it says)
2. **LLM conversation log** — Share a link to the sessions or invite osama@bu.edu to it. Also, Export or screenshots of your key sessions. I want to see *how* you worked with the tool: your prompting strategy, where you pushed back, where you caught errors.
3. **One-page reflection** answering:
    - What did the LLM get wrong, and how did you catch it?
    - What prompting approaches worked best?
    - What did you learn that the LLM couldn't have told you directly?

- Which 1–2 project ideas emerged that you might pursue in later phases?

Grading (100 pts)

Component	Points	What I'm looking for
Technical accuracy	30	Claims verified against real papers; no hallucinated references
Tutorial quality	25	Clear, well-structured, teaches from first principles, good examples
Quiz integration	15	Questions test understanding (not trivia), distributed through the tutorial, with explanations
LLM collaboration	15	Evidence of iterative, critical use — not one-shot copy-paste
Reflection	10	Honest, specific, shows critical thinking about the tool
References	5	Real, relevant, correctly cited

Ground Rules

- **Using the LLM is required, not cheating — for this assignment.** The skill being graded is *directing* the tool and *verifying* its output. Blind copy-paste will be obvious and will score poorly on accuracy.
- Ask the LLM to quiz *you* while you read the papers — it's a good way to check your own understanding before you write.
- Cite the papers, not the LLM. The LLM is your drafting partner, not a source.

Suggested Workflow (7 days)

- **Days 1–2:** Pick area, skim candidate papers, select your 3–5, read them properly.
- **Days 3–4:** Outline the tutorial yourself first, then iterate with the LLM section by section. Generate quiz questions per section as you go.
- **Day 5:** Verification pass — check every claim and reference against the sources.
- **Day 6:** Polish, formatting, answer keys.
- **Day 7:** Write the reflection, package the conversation log, submit.